

# Printing Receipts, and Getting the Register onto the App Store

For Michael. Written to be followed start to finish, with nothing assumed.

This picks up where *Getting the Register onto Your iPad* (SLICE 9) left off. That document got the app onto one iPad through a cable. This one makes the receipt printer work from inside the app, and gets the app onto the App Store so it can be installed normally on every till.

There is one thing in here with a **1 to 2 week waiting period**, and if you miss it your App Store submission gets rejected. It is Part 5. Please read Part 5 early even if you do everything else later.

## Read this part first (3 minutes)

### What changed in the app

Before this week, when a cashier tapped **Print**, the iPad jumped out of the register and into Star's PassPRNT app. The receipt printed, and the cashier was left staring at the wrong app with a customer in front of them. There were **seven** places in the register that did this.

They are all gone. The register now talks to the printer itself. The screen never changes.

### The fact that changed how I built it

I want to show you this one, because it is a good example of why the "never guess" rule earns its keep.

Your printer is a **TSP143IIIBi**, which is part of Star's **TSP100III** family. Star's programming manual says this, in a note that is easy to miss:

*TSP100III series and TSP100IIU+ do not support `actionPrintText` because these products are **graphics-only printers**. Please use the `actionPrintImage` method.*

In plain English: **your printer cannot be sent text**. It can only be sent pictures. From memory, I would have sent it text — that is how nearly every other receipt printer works. The code would have compiled. It would have passed Apple's review. It would have printed **absolutely nothing**, and you would have found out at the counter.

So the app now draws your receipt on an invisible page, takes a picture of it, and sends the picture. Slightly more work, and the only thing that actually prints on this hardware.

### Your two printers — you were right

You said the equipment page already had it. It does. There are two Star printers in there and they do different jobs:

| Tag            | Model                                           | Job                                                                                                |
|----------------|-------------------------------------------------|----------------------------------------------------------------------------------------------------|
| PRN-COUNTER-01 | TSP143IIIBi, Bluetooth, serial 2550923021300119 | The <b>front counter</b> printer. Register sales, and it kicks the cash drawer. <b>This slice.</b> |
| PRN-RECEIPT-01 | TSP143IV, Ethernet/CloudPRNT                    | Auto-prints <b>online</b> orders. Different printer, different job, untouched.                     |

The app is written so it can only ever talk to PRN-COUNTER-01 . There is a test that fails if anyone ever confuses the two.

### When the printer breaks

The rule I built to, written at the top of the printing code:

**A cash sale is never blocked by a printer.**

Out of paper, printer off, Bluetooth dropped, drawer jammed — the sale still completes and is still recorded. You get a plain-English message like *"The printer is out of paper"* rather than a spinning wheel. If the drawer did not open, it tells you so and reminds you that **No Sale** will open it.

## What you need in front of you

- Your MacBook Pro, plugged in
- The register iPad (the 12.9-inch Pro, serial DLXQNCCEMGMW3)
- The Star TSP143IIBi, plugged into power, with paper in it
- Your Apple Developer account login
- About 90 minutes of hands-on time, plus the 1–2 week MFi wait in Part 5

## The address question — my recommendation

You asked whether to keep using the Vercel site for testing and switch to GreenwayMarijuana.com at launch, and you asked me to decide.

**Yes. Keep Vercel for testing, switch at launch.** Here is why, and I checked all three addresses again today rather than trusting last week's notes:

| Address                                          | What it does today                                                                                 |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <code>https://greenwaymarijuana.com</code>       | Loads fine, valid certificate — but it is your <b>old WordPress site</b> . It has no register API. |
| <code>https://www.greenwaymarijuana.com</code>   | <b>Does not load at all.</b> The certificate does not cover <code>www</code> .                     |
| <code>https://greenwaywebsite1.vercel.app</code> | The <b>new site</b> . Register API answers correctly.                                              |

I owe you a correction here: in the SLICE 9 document I described the apex domain as if it were not the real public site. You corrected me, and you were right — `greenwaymarijuana.com` **is** your live public site today, and the Vercel address is the not-yet-public build of its replacement.

**So: build the app pointing at `https://greenwaywebsite1.vercel.app`.**

That is the right choice for a specific reason beyond convenience. The register needs a server that has the new API on it. Your apex domain currently serves WordPress, so pointing registers at it today would simply break them. And you want to test before you deploy — which is exactly what the Vercel address is for.

**Changing it later is not a big deal.** It is one line and a rebuild, about ninety seconds of work plus a new build to the iPads. Nothing is locked in, and you are not committing to anything by using Vercel now.

**The switch at launch, when you are ready,** is three steps in this order:

1. Point `greenwaymarijuana.com` (and `www`) at the new site instead of WordPress, and let the certificate issue for **both** names.
2. Confirm `https://www.greenwaymarijuana.com` loads without a warning. Right now it does not, and that is worth fixing regardless of the register — customers who type `www` today get a security warning on your business.
3. Rebuild the register app with the new address and push the update.

One caution worth stating plainly: **do not switch the domain and the app on the same day you are busy.** Do it on a slow morning, with the Vercel address still working as a fallback.

## Part 1 — Add Star's printer software to the project (Mac, ~15 min)

The Swift code that drives the printer is written and committed. What is not committed is Star's own SDK, because it is their software and it is downloaded by Xcode on demand. You do this once.

### Step 1 — Get the latest code

Open **Terminal** on the Mac and run these one at a time:

```
cd ~/greenway/GREENWAY-WEBSITE
git pull
npm ci
```

`npm ci` takes a few minutes and prints a lot. That is normal.

## Step 2 — Open the project in Xcode

⚠ **CORRECTED.** An earlier version of this step listed `npm run register:build` with **no server address**. That builds a register that installs, launches, and then refuses to start with "This register cannot start." Use the command below instead. **My mistake, not yours.**

```
REGISTER_API_BASE="https://greenwaywebsite1.vercel.app" npm run register:build:ios  
npx cap open ios
```

Two things changed from the old version and both matter:

- `register:build:ios` instead of `register:build` — the `:ios` one runs the **preflight check** and `cap sync ios` for you. The plain one skips both.
- The `REGISTER_API_BASE=` prefix — this is the server address, and it is baked into the bundle at build time. Without it the app has no idea what to call.

You should see this line go past:

```
OK    server address: The app will call https://greenwaywebsite1.vercel.app/api/pos/... for every request.
```

If instead it stops with `FAIL server address`, the address was not passed — re-run the command with the whole `REGISTER_API_BASE="..."` prefix on the same line.

Xcode opens. If it starts indexing, let it finish before clicking anything.

*If `npx cap` complains about your Node version, run `node --version`. You need **22 or newer**. Install it from [nodejs.org](https://nodejs.org), close Terminal, open a new one, and try again. (My sandbox is stuck on Node 20, which is why I could not run these commands for you.)*

## Step 3 — Add the StarXpand SDK

In Xcode:

- Menu bar → **File** → **Add Package Dependencies...**
- In the search box at the top right, paste exactly:  

```
https://github.com/star-micronics/StarXpand-SDK-ios
```
- Press Return. A package called **StarXpand-SDK-iOS** appears.
- Leave **Dependency Rule** at *Up to Next Major Version*.
- Click **Add Package**. Xcode downloads it — takes a minute or two.
- A second window asks which target to add it to. Make sure **StarIO10** is checked and the target next to it says **App**. Click **Add Package**.

## Step 4 — Check it took

In the left sidebar of Xcode, scroll to the bottom. You should see **Package Dependencies** with **StarIO10** under it, with a version number next to it (for example `2.13.0`). Alongside it you will also see `CapApp-SPM local` and `capacitor-swift-pm 8.5.0` — those two are ours and were already there.

Seeing **StarIO10** in that list means Xcode successfully *downloaded* the SDK.

## Step 4b — Check the SDK is actually attached to the app

This is a **different thing** from Step 4, and it is the step people miss.

Step 4 proves Xcode downloaded the SDK. It does **not** prove the SDK is attached to our app. A package can sit in that sidebar list, fully downloaded, and still not be wired into the app — in which case the build fails with:

```
No such module 'StarIO10'
```

To check:

- In the left sidebar, click the blue **App** icon at the very top.
- In the main panel, select the **App** target (left column, under TARGETS — not the one under PROJECT).

3. Click the **General** tab.
4. Scroll down to **Frameworks, Libraries, and Embedded Content**.

**StarIO10** must be listed there.

- **If it is listed** — you are done, move to Step 5.
- **If the list is empty, or StarIO10 is missing** — click the + button underneath the list. A picker opens. Type `StarIO10` in the search box, select **StarIO10**, and click **Add**. It now appears in the list.

Then move to Step 5.

## Step 5 — Build it

Press **Cmd-B**. This is the first time this Swift code has ever been compiled — I have no Swift compiler in my sandbox, so I could not do it for you. I wrote it line by line against Star's published manual, but I am telling you plainly that **it has never been compiled**, so a small error is possible.

- **"Build Succeeded"** → carry on to Part 2.
- **Red errors** → screenshot the errors and send them to me. Do not try to fix them yourself. Errors here will be small and mechanical, and I would rather fix them correctly than have you guess.

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## Part 2 — Pair the printer to the iPad (~10 min)

This is Bluetooth pairing at the iPad level. It is separate from the app, and you do it once per iPad.

### Step 6 — Find the printer's name

1. Turn the printer **off** using the switch on its side.
2. Hold down the **FEED** button on the front, and while holding it, turn the printer **on**.
3. Keep holding until it starts printing, then let go.

It prints a test page. Find the line that says **Dev Name**. It looks like `TSP100-XXXXX`. **Write it down** — you need it in the next step and again in Part 5.

### Step 7 — Pair it

On the **iPad**:

1. **Settings** → **Bluetooth**
2. Make sure Bluetooth is **On**
3. Under *Other Devices*, find the `TSP100-XXXXX` name from Step 6
4. Tap it
5. Wait for it to say **Connected**

***If the printer does not appear:** confirm it is on and that the blue light is steady rather than flashing. If it still does not show, hold the small **RST** button on the back while switching the printer on, keep holding through two reset beeps, then release, turn it off and on, and try again. That is Star's factory reset for Bluetooth.*

### Step 8 — Lock it down (recommended)

Star recommends turning off **New Pairing Permission** once you are paired, so nobody in the parking lot can pair to your printer. Install **Star Quick Setup Utility** from the App Store, open **Printer Settings** → **Bluetooth Settings**, turn **New Pairing Permission** to **OFF**, and tap **Apply**.

***Read this before you do it.** With that setting off, you cannot pair a new iPad without factory-resetting the printer first. If you plan to add a second till soon, **pair both iPads first**, then turn this off.*

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## Part 3 — Test printing on the real iPad (~10 min)

### Step 9 — Install the new build

With the iPad connected to the Mac by cable, in Xcode choose your iPad from the device menu at the top and press the ► **Play** button. Same as SLICE 9.

### Step 10 — Run a test sale

1. Open the register on the iPad
2. Ring up any item and complete a **cash** sale
3. Watch what happens

**What should happen:** the receipt prints, the drawer pops, and **the screen never leaves the register**. No jump to another app. That is the whole point of this slice.

### Step 11 — Test the drawer rules

These are deliberate and worth confirming:

| Do this              | Drawer should    |
|----------------------|------------------|
| Cash sale            | <b>Open</b>      |
| Refund               | <b>Open</b>      |
| Void                 | <b>Open</b>      |
| No Sale              | <b>Open</b>      |
| Reprint a receipt    | <b>Stay shut</b> |
| Print the day report | <b>Stay shut</b> |

Reprint not opening the drawer is intentional — otherwise anyone could pop the till by reprinting an old receipt.

### Step 12 — Test it failing

Please actually do this one. Open the printer's paper lid so it cannot print, and ring up a cash sale.

**Expected:** the sale still completes and is recorded, and you get a clear message about the printer. **The sale must not be blocked.** If a printer problem ever stops a sale from completing, that is a bug — tell me immediately.

## Part 4 — What is not built yet

✓ **RESOLVED IN SLICE 11 — see** `MICHAEL-slice11-printer-picker.md`

The printer picker described below has been built and merged (PR #1057). It is on the iPad at **MORE ► ↗ Receipt printer**.

Building it also uncovered **two bugs in this slice** that would have stopped pairing from working at all, even once the screen existed: the app rejected Bluetooth printer IDs because it expected a MAC address (Bluetooth uses the printer's name), and the chosen printer was never actually saved. Both are fixed and explained in Parts 4 and 5 of the SLICE 11 doc.

The rest of this section is left as written, for the record.

Being straight with you about a gap.

There is no **printer settings screen** in the app yet — no screen listing nearby printers for you to pick one. The plumbing is written and tested ( `discoverPrinters` , `setPairedPrinterIdentifier` ), but nothing calls it.

Until that screen exists, the app has no saved printer, so it falls back to the old PassPRNT route instead of failing a sale. **This means Part 3 may still bounce to PassPRNT**, and that is expected right now, not a failure.

This is one small slice of work. Say the word and it is the next thing I build. I did not bundle it in here because this slice was already large, and I would rather ship the printing engine verified than ship two half-things.

# Part 5 — MFi approval ⚠️ START THIS EARLY

This is the part with the waiting period. Please read it now even if you do the rest later.

## Why this exists

Your printer connects over **Bluetooth Classic**, and Apple requires any accessory like that to be part of its **MFi** program. Your app has to be registered against the printer before Apple will approve it.

Here is the trap: **you do not apply to Apple. Star does, on your behalf.** If you submit your app to the App Store without this, Apple **rejects it**, and you have to wait out the 1-2 weeks anyway. Doing it now costs you nothing.

Straight from Star's own page:

*In order to offer your app on the Apple App Store, your app needs to be approved by the Apple MFi program **before you submit it** to the Apple App Store.*

## Step 13 — Submit Star's form

Go to:

`https://star-m.jp/eng/support/s_print/app_regist.html`

Fill it in. Here is **exactly** what it asks and what to put:

### Developer Information

| Field                                  | What to enter                                       |
|----------------------------------------|-----------------------------------------------------|
| Name                                   | Michael Lyman                                       |
| E-mail address                         | The one you check. Star replies here with the PPID. |
| E-mail (confirm)                       | Same                                                |
| Company name                           | Greenway Marijuana                                  |
| Area                                   | <b>US &amp; Canada</b>                              |
| Address / City / State / Country / Zip | Your Port Orchard, WA business address              |
| Telephone                              | Include the country code: <code>+1</code>           |

### Application Information

| Field                                      | What to enter                                         |
|--------------------------------------------|-------------------------------------------------------|
| Printer(s) your iOS APP communicates with  | Tick <b>TSP100IIIBI</b> (under <i>TSP100 Series</i> ) |
| Name of app as it will appear in App Store | <code>Greenway Point of Transaction</code>            |
| APP version                                | <code>1.0</code>                                      |
| Bundle Identifier                          | <code>com.greenwaymarijuana.register</code>           |
| Description of your app                    | See the paragraph below                               |

⚠️ **Tick TSP100IIIBI, not TSP143IV.** *TSP143IV is your online order printer, on Ethernet. The one this app talks to is the Bluetooth TSP100IIIBI.*

**Description** — the form says single-byte characters only, so plain English, no fancy punctuation. You can paste this:

`A point of sale register application for a licensed cannabis retail store in Washington State. Staff ring up in-store sales on an iPad, take payment, and print a customer receipt on a Star TSP143IIIBi over Bluetooth. The app also opens the cash drawer connected to the printer DK port after a cash sale, a refund, a void, or a no sale. The app is for internal store staff use only.`

Tick the privacy policy box and click **Confirm**, then submit.

Step 14 — Wait for the PPID

Star files it with Apple, then emails you an **MFi Product Plan ID (PPID)**. It looks like `MFi PPID 123456-7890` .

**Expect 1 to 2 weeks.** When it arrives, **save it somewhere you will find it again** — you need it in Step 20, and you will need it for every future version of the app.

If two weeks pass with no word, chase them through the contact form at `https://star-m.jp/eng/support/s_print/` .

Part 6 — Putting the app on the App Store

Do this once the PPID has arrived.

A decision worth making first

There are two ways to get this app onto your iPads, and the App Store is not obviously the right one:

| Route                               | What it means                                                                                                                    |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>App Store (private/unlisted)</b> | Install from the App Store, updates arrive automatically. Requires MFi and Apple review of every version.                        |
| <b>Ad Hoc / cable install</b>       | What you did in SLICE 9. No review, no MFi, no waiting — but you plug in each iPad for every update, and builds expire annually. |

You asked how to get it on the App Store, so that is what is written below. But for two tills in one store, honestly, the cable route is often less friction. The App Store route wins when you stop wanting to touch iPads by hand.

Either way, **do Part 5 now** — it is free and it is the long pole.

Step 15 — Create the app record

On the **Mac**, go to `https://appstoreconnect.apple.com` and sign in.

- 1. **My Apps** → the **+** → **New App**
- 2. Platform: **iOS**
- 3. Name: `Greenway Point of Transaction`
- 4. Primary language: **English (U.S.)**
- 5. Bundle ID: pick **com.greenwaymarijuana.register** from the dropdown
- 6. SKU: `GREENWAY-POT-1` (internal only, never shown)
- 7. User Access: **Full Access**
- 8. **Create**

*If the bundle ID is not in the dropdown, it has not been registered yet. Go to `developer.apple.com` → **Certificates, Identifiers & Profiles** → **Identifiers** → **+** → **App IDs** → **App**, enter the description `Greenway Point of Transaction` and the bundle ID `com.greenwaymarijuana.register` , and register it. Then come back.*

Step 16 — Set the version to Distribution

In Xcode: click **App** in the left sidebar → **Signing & Capabilities** → confirm **Automatically manage signing** is ticked and your **Team** is selected.

Step 17 — Archive

- 1. In the device menu at the top, choose **Any iOS Device (arm64)** — *not* your iPad, and *not* a simulator. Archive is greyed out otherwise.

⚠ **This setting is ONLY for archiving, in this step.** If you leave it on **Any iOS Device (arm64)** and press ▶ *Play*, the build fails with "A build only device cannot be used to run this target" — and, because Xcode then has no real device registered, it also fails signing with "Your team has no devices from which to generate a provisioning profile." Both errors are the same cause. Switch the destination back to **your iPad by name** before you press *Play* again. See *Troubleshooting the build* at the end of this document.

2. Menu bar → **Product** → **Archive**
3. Wait. Five to ten minutes is normal.

The **Organizer** window opens with your archive listed.

### Step 18 — Upload

1. In Organizer, select the archive → **Distribute App**
2. Choose **App Store Connect** → **Next**
3. Choose **Upload** → **Next**
4. Accept the defaults on the following screens → **Next**
5. **Upload**

Then wait again. The build shows in App Store Connect as *Processing* for 10 to 60 minutes. You get an email when it is ready.

### Step 19 — Fill in the listing

Back in App Store Connect, in your app:

- **Description** — what the app does
- **Keywords**
- **Support URL** — `https://greenwaymarijuana.com`
- **Screenshots** — required, for 12.9-inch iPad Pro. Take them on the iPad with the top button + volume up, and AirDrop them to the Mac.
- **App Privacy** — you must complete this section or you cannot submit
- **Age Rating** — answer honestly; this is a cannabis retail tool
- **Pricing** — Free

### Step 20 — ⚠ The PPID. Do not skip this.

**This is the step that gets apps rejected.**

In your app in App Store Connect, scroll down to **App Review Information**. In the **Notes** box, type your PPID from Step 14, exactly like this:

MFI PPID 123456-7890

(with your real number, obviously)

Star's instruction, word for word:

*Submit your App to the Apple App Store through App Store Connect with the MFI PPID number in the Notes section of the App Review Information.*

While you are in that box, it is worth adding a note for the reviewer, because this app is useless to them without a login and a printer:

This is an internal point of sale application for a single licensed cannabis retailer in Washington State. It requires a store-issued device credential to operate and connects to a Star TSP143IIIBi Bluetooth receipt printer.  
MFI PPID 123456-7890  
Demo credentials are provided below.

Then fill in the demo account boxes with a test login. **Reviewers will reject an app they cannot get into.**

### Step 21 — Submit

Select your processed build, click **Add for Review**, then **Submit for Review**.

Review usually takes a day or two. Star notes the whole approval can run 1 to 2 weeks including their part.

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### If something goes wrong

| What you see                  | What it means                                | What to do                                                               |
|-------------------------------|----------------------------------------------|--------------------------------------------------------------------------|
| Xcode red errors after Step 5 | My uncompiled Swift has a mistake            | Screenshot and send to me. Do not guess.                                 |
| Printer not in Bluetooth list | Not on, or already paired elsewhere          | Factory reset: hold <b>RST</b> while switching on, through two beeps     |
| Receipt prints blank          | Wrong print mode for a graphics-only printer | Tell me — this is the <code>actionPrintImage</code> issue and it is mine |
| Prints, but drawer stays shut | Drawer cable, or a reprint (correct)         | Check the DK cable; try <b>No Sale</b>                                   |
| Still jumps to PassPRNT       | No printer paired in-app yet                 | Expected — see Part 4                                                    |
| Apple rejects: MFi/PPID       | PPID missing from review notes               | Step 20                                                                  |
| Sale blocked by printer error | <b>A real bug</b>                            | Tell me immediately. This must never happen.                             |
|                               |                                              |                                                                          |

## What was verified, not assumed

Everything above was checked against a primary source today. Specifically:

- The **graphics-only** limitation is from Star's StarXpand manual, not memory. This is the fact that would have silently broken your receipts.
- The **MFi process, the 1-2 week window, and the PPID going in App Review Notes** are quoted from Star's own MFi page.
- The **registration form fields** were read off the live form. `TSP100IIIBI` is listed there as a real option.
- The **pairing steps** and the **RST reset** are from Star's TSP100IIIBI online manual.
- The **three web addresses** were tested today: the apex serves WordPress with a valid certificate, `www` fails its certificate check, and the Vercel site serves the new app and answers the register API.
- Both printers were read out of the database migration in your own repo.

And what was **not** verified, stated plainly:

- **The Swift plugin has never been compiled.** No Swift toolchain here. First real compile is Step 5 on your Mac.
- **Nothing has been printed on real paper.** No printer here either.
- The printer settings screen does not exist yet (Part 4).

## Troubleshooting the build

### "A build only device cannot be used to run this target"

### + "Your team has no devices from which to generate a provisioning profile"

**These two errors are one problem, and the fix takes five seconds.**

They are **not** a code problem, **not** an Apple Developer account problem, and **not** a problem with your signing setup. Nothing is wrong with your account, your team, or the app. Your team shows correctly as *LYMAN'S MARIJUANA L.L.C.* and *Automatically manage signing* is ticked, which is exactly right.

**The cause:** the destination at the top of the Xcode window is set to **Any iOS Device (arm64)**.

That is not a real device. It is a placeholder that means "*build something generic for later distribution*" — it is what you select to make **Archive** work in Step 17. You cannot **run** on it, because there is no device there to run on. Hence "*a build only device cannot be used to run this target.*"

The signing error is a **knock-on effect of the same thing**, not a second fault. With no real device selected, Xcode has no device to write into a development provisioning profile, so it asks Apple for a profile covering zero devices and Apple declines.

That is what "*your team has no devices from which to generate a provisioning profile*" means. It is describing the empty destination, not a broken account.

## The fix

1. At the **top centre** of Xcode, click where it says **Any iOS Device (arm64)**.
2. In the dropdown, under a heading like **iOS Device**, choose **your iPad by name**.
3. Press ► **Play**.

Both errors clear together. You do not need to press *Try Again*, sign in again, register a device by hand at developer.apple.com, or change the bundle identifier.

## If your iPad is not in that dropdown

Work down this list in order — the first two are the usual answer:

1. **Unlock the iPad** and leave it on the Home screen. A locked iPad often does not appear.
2. **Cable.** Use a real data cable. A charge-only cable is the single most common cause, and it looks identical to a good one.
3. **Trust.** Unplug, plug back in, and on the iPad tap **Trust This Computer**, then enter the passcode.
4. **Developer Mode** must be on: iPad **Settings ► Privacy & Security ► Developer Mode ► On**, then restart the iPad. This only appears after the iPad has been plugged into a Mac running Xcode. (SLICE 9 doc, Step 9.)
5. **Wait 30 seconds** after plugging in. Xcode has to prepare the device the first time, and it is silent while it does.

## Why my instructions sent you here — my fault, not yours

Step 17 of this document tells you to select **Any iOS Device (arm64)**, and it is right to, because Archive genuinely requires it. What Step 17 did **not** say is that the setting must be changed **back** to your iPad before pressing Play. If you had been anywhere near Step 17, or had it selected for any other reason, this failure was the guaranteed next thing to happen — and the signing error on top of it makes it look far more serious than it is.

I have added a warning to Step 17 so the next person through does not hit it.

## What this error does NOT mean

Worth saying plainly, because the wording is alarming:

- It does **not** mean your Apple Developer account is wrong or unapproved.
- It does **not** mean the £99/\$99 membership has a problem.
- It does **not** mean you need to add your iPad's UDID manually anywhere.
- It does **not** mean the Swift printer plugin failed to compile.

On that last point: **this failure happened before the compiler ran**. So it tells us nothing yet about whether my printer code builds. Once you select your iPad and press Play, watch for what comes next:

- **"Build Succeeded"** then the app launches on the iPad → the Swift plugin compiled. Carry on to Part 2 / Part 3.
- **Red errors naming a `.swift` file** → that is the real first compile of my plugin. Screenshot them and send them to me. Do not fix them yourself.

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# Troubleshooting: "No profiles for 'com.greenwaymarijuana.register' were found"

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If Cmd-B fails with **both** of these:

```
No profiles for 'com.greenwaymarijuana.register' were found
Xcode couldn't find any iOS App Development provisioning profiles
matching 'com.greenwaymarijuana.register'.
```

```
Communication with Apple failed
Your team has no devices from which to generate a provisioning profile.
Connect a device to use or manually add device IDs in Certificates,
```

...then this section is for you. **This is not a code problem.** Nothing is wrong with the app, the printer SDK, or the Swift plugin. This is Apple's code-signing paperwork, and it has to be done once per Apple ID.

## What is actually going on

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Apple will not let *any* app run on *any* real iPad unless Apple has personally blessed that exact combination of:

1. **A developer identity** — proof of who you are (your Apple ID / team)
2. **An App ID** — the app's unique name, `com.greenwaymarijuana.register`
3. **A registered device** — your specific iPad, by its hardware serial (UDID)

Those three get bundled into a **provisioning profile**. Xcode's "Automatic signing" will create that profile *for* you — but it cannot, because right now Xcode does not know who your team is. The project file has **no DEVELOPMENT\_TEAM set at all**. I checked; the setting does not appear anywhere in the project.

The second error is a direct consequence of the first. Apple is saying "your team has no devices" because Xcode has not yet told Apple which team you are, so it is looking at an empty team.

**Both errors are one missing setting.** Fix it once and both disappear.

## Fix it — click by click

---

### Step A — Make sure your Apple ID is in Xcode

1. Xcode menu bar → **Xcode** → **Settings...** (older Xcode: *Preferences...*)
2. Click the **Accounts** tab at the top.
3. Look at the left-hand list. Is your Apple ID there?
  - **Yes** → good, close this window, go to Step B.
  - **No** → click the **+** at the bottom left → choose **Apple ID** → **Continue** → sign in with the Apple ID that has the \$99 Apple Developer membership. Approve the two-factor prompt on your iPhone or iPad.
4. Click your Apple ID in the list. On the right you should see a team — likely **Michael Lyman (Individual)** or your business name. If you see a team listed, you are good.

Close Settings.

### Step B — Tell the project which team to use

**This is the actual fix.**

1. In the left sidebar of Xcode, click the blue **App** icon at the very top.
2. In the middle panel, under **TARGETS**, click **App**. (Not the one under PROJECT — the one under **TARGETS**.)
3. Click the **Signing & Capabilities** tab along the top. (It is right next to *General*, which is the tab you were just on.)
4. Make sure **Automatically manage signing** is **checked** ✓
5. Find the **Team** dropdown. It almost certainly says **None**.
6. Click it and select your team — **Michael Lyman (Individual)** or whatever your developer account is named.

Now watch that panel for about 10–30 seconds. Xcode talks to Apple, registers your iPad, creates the App ID, and generates the profile automatically. The red error text in that panel should replace itself with a line reading something like "*Provisioning Profile: Xcode Managed Profile.*"

### Step C — Your iPad must be plugged in and trusted

The second error — "*your team has no devices*" — means Apple has never seen your iPad. Xcode registers it automatically, but **only if the iPad is connected and trusted at that moment**.

1. Keep the iPad plugged into the MacBook with the cable.
2. Unlock the iPad — actually enter the passcode so you are on the home screen.
3. If the iPad shows "**Trust This Computer?**", tap **Trust** and enter the passcode.

4. At the top of the Xcode window, click the destination name and select **your iPad by name**.

△ Your screenshot showed this set to **"Any iOS Device (arm64)"** — that is a placeholder, not a real device, and it is part of why Apple says you have no devices. **Change it to your iPad by name**. See the previous troubleshooting section for the full explanation of that trap.

5. Go back to **Signing & Capabilities** and confirm the errors are gone. If they are still showing, toggle **Automatically manage signing** off and back on to force Xcode to retry.

## Step D — Build

Press **Cmd-B**.

## If it still fails

---

**"Failed to register bundle identifier"** — the ID `com.greenwaymarijuana.register` is already taken by another Apple account. Tell me and I will change it; it is a one-line change on my side.

**"Unable to log in with account"** — your Apple Developer membership may not be fully active. Go to <https://developer.apple.com/account/> and sign in. If it asks you to accept a new legal agreement, accept it, then retry Step B. This is a very common cause and Apple gives no useful hint about it.

**Team dropdown is empty / only shows "None"** — your Apple ID is signed in but has no developer membership attached. Check <https://developer.apple.com/account/> shows an active membership.

**"Personal development teams do not support Push Notifications"** or similar capability complaints — we use no special capabilities, so this should not appear. Send me a screenshot if it does.

## What this does NOT mean

---

- It does **not** mean the printer SDK failed. Your screenshot confirms **StarIO10** is correctly listed under *Frameworks, Libraries, and Embedded Content* alongside *CapApp-SPM*. That part is done and correct.
- It does **not** mean my Swift code is broken. **Signing happens before compiling**. The compiler still has not run, so we still do not know whether the printer plugin builds.

Once signing is fixed, the next build is the first real compile of `StarPrinterPlugin.swift`. If red errors appear naming a `.swift` file and a line number — screenshot them and send them to me. Do not try to fix them yourself.

# Troubleshooting: "0 Provisioned Devices" — the iPad was never registered

---

If you have set the **Team** and the *"Your team has no devices"* error is **still** showing, this section is for you. You are close. Setting the team was the right move and it worked. There is one thing left.

## How to confirm this is your problem

---

Xcode → **Settings...** → **Accounts** → click your team. Look at **On Device Testing**.

If it says **0 Provisioned Devices**, Apple has never been told your iPad exists. That is the entire remaining problem.

## Why this happens

---

Xcode registers your iPad with Apple **automatically** — but only at a very specific moment: when it needs to build **for that exact iPad**.

If the destination at the top of the Xcode window says **"Any iOS Device (arm64)"**, that moment never arrives. That is a placeholder, not a real device. Xcode has no serial number to send to Apple, so it registers nothing, so your team stays at 0 devices, so no profile can be generated.

This is the same "Any iOS Device (arm64)" trap from the earlier troubleshooting section, showing up wearing a different mask. The first time it blocked the Run button. This time it is blocking device registration.

## Which team to pick

If you have two teams in the dropdown, they are not equal:

| What you see in Accounts                                                                       | What it means                                                                |
|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| <b>Developer Team</b> + Role: Admin + green ✓ <i>Certificates, Identifiers, &amp; Profiles</i> | This is the <b>paid</b> \$99 team. <b>Use this one.</b>                      |
| <b>Personal Team</b>                                                                           | The free tier. Apps expire after 7 days and it cannot ship to the App Store. |

Pick the **paid Developer Team**. The Personal Team will technically work for a quick test, but the app stops running after 7 days and you cannot submit with it, so there is no reason to use it.

## Fix it — in this exact order

**The order matters.** Selecting the iPad must happen *before* Xcode can register it.

1. **Plug the iPad into the MacBook** with the cable.
2. **Unlock the iPad.** Actually type the passcode and get to the home screen. A locked iPad is invisible to Xcode.
3. If "**Trust This Computer?**" appears on the iPad → tap **Trust** → enter the passcode. If it does not appear, unplug and replug the cable.
4. In Xcode, click the **destination** at the top of the window (the part that currently reads *Any iOS Device (arm64)*).
5. Look for your iPad **by name** near the top of that list, under a heading that says something like *iOS Device*.
  - **It is there** → click it. Go to step 6.
  - **It is not there** → see *If the iPad does not appear* below.
6. Go to **Signing & Capabilities** → set **Team** to your paid **Developer Team**.
7. Click **Try Again** next to the red error.

Wait 10–30 seconds. Xcode sends the iPad's serial number to Apple, registers it, creates the App ID, and generates the profile. The red errors should disappear and be replaced by a normal *Provisioning Profile: Xcode Managed Profile* line with no error underneath.

8. Press **Cmd-B**.

To confirm it worked: Xcode → Settings → Accounts → your team → **On Device Testing** should now read **1 Provisioned Device**.

## If the iPad does not appear in the destination list

1. Xcode menu bar → **Window → Devices and Simulators**
2. Your iPad should be listed on the left. Click it.
3. If it says "**Preparing debugger support...**" or "**Waiting to reconnect**", wait. On a first connection this can take 10–20 minutes. Do not unplug it.
4. If it shows a "**Trust**" prompt or "*Unlock the device*", do that on the iPad itself.
5. Try a different cable, and use a port directly on the MacBook rather than through a hub or dock. A charge-only cable will charge the iPad but carry no data — this is a very common cause and looks identical to a broken iPad.

## Fallback: register the iPad by hand

If the automatic route keeps failing, you can register the iPad manually. You are an **Admin** on the LLC team, so you have permission to do this.

### Get the iPad's identifier (UDID):

1. Xcode → **Window → Devices and Simulators**

2. Click your iPad on the left
3. Near the top you will see **Identifier** followed by a long string of letters, numbers and dashes. Right-click it → **Copy**.

### Register it with Apple:

1. Go to <https://developer.apple.com/account/resources/devices/list>
2. Sign in with the account that owns the **paid** team
3. Click the + (Register a New Device)
4. **Platform:** iOS, iPadOS, tvOS, watchOS
5. **Device Name:** Greenway Counter iPad (any name you like)
6. **Device ID (UDID):** paste what you copied
7. Click **Continue** → **Register**

Then back in Xcode: **Signing & Capabilities** → toggle **Automatically manage signing** off and back on. It will now find the device.

## Still nothing?

Check <https://developer.apple.com/account/> for a banner asking you to accept an updated legal agreement. If one is waiting, **nothing** device-related will work until you accept it, and Apple gives no useful hint that this is the cause. Accept it, then retry.

## Reminder

Signing still happens **before** compiling. The compiler has not run yet. Once the build gets past signing, watch for red errors naming a `.swift` file and a line number — that is the first real compile of the printer plugin. Screenshot those and send them to me rather than trying to fix them.

# Troubleshooting: "iPad (Developer Mode disabled)" / enable Developer Mode

**This is real progress, not a new problem.** Getting this message means the iPad registration from the previous section **worked** — Xcode can now see your iPad by name in the destination list. This is a normal, expected, one-time step that every iPad goes through the first time it's used for development. It is not related to signing, teams, or the printer SDK at all.

## Why this exists

Apple deliberately makes "run code that isn't from the App Store" an opt-in, visible decision on the device itself — not something a computer can turn on by remote control. This protects you from someone plugging in a cable and silently installing something. You have to physically confirm it, on the iPad, on purpose.

## Turn it on — do this ON THE IPAD

1. On the iPad: **Settings** → **Privacy & Security**
2. Scroll down. Near the bottom, find **Developer Mode**. Tap it.
3. Turn the switch **on**.
4. The iPad will warn you that Developer Mode reduces security and offer a **Restart** button. Tap **Restart**. This is expected — let it restart.
5. **After the iPad turns back on and you unlock it**, a second alert appears confirming you want to enable Developer Mode. This step is easy to miss — it does not happen automatically just because you restarted.
6. Tap **Enable**.
7. Enter your iPad's passcode to confirm.

Developer Mode is now on.

## Back in Xcode

1. Keep the iPad plugged into the MacBook.
2. Click the destination dropdown at the top of the Xcode window again.
3. It should now read **iPad** without the "(Developer Mode disabled)" label. If it still shows the disabled label, unplug and replug the cable.
4. Confirm your iPad is selected (not "Any iOS Device (arm64)" — see the earlier section if it reverted).
5. Press **Cmd-B**.

## If Developer Mode doesn't appear in Settings at all

---

Per Apple's own documentation, Developer Mode only appears in **Privacy & Security** once the iPad has started the pairing process with your Mac at least once. If you don't see it:

1. Make sure the iPad is plugged in, unlocked, and you tapped **Trust** on the "Trust This Computer?" prompt earlier.
2. In Xcode, open **Window → Devices and Simulators**, and confirm your iPad is listed there. Just having it appear in that window is usually enough to trigger the Developer Mode toggle to show up in Settings within a minute or two.
3. If it still doesn't appear, unplug the iPad, lock it, unlock it, and plug it back in.

## What this does NOT mean

---

- It does not mean anything is wrong with your Apple Developer account.
- It does not mean the provisioning profile or team setup from the earlier sections was wrong — quite the opposite, it means that part is now working well enough that Xcode is trying to talk to your specific iPad.
- It does not mean the printer SDK or my Swift code has a problem.

## Reminder

---

Signing and device trust both still happen **before** compiling. Once Developer Mode is on and the build gets past this point, watch for what happens next:

- **"Build Succeeded"** and the app launches on the iPad → the printer plugin compiled. Move on to Part 2 / Part 3 of this guide.
- **Red errors naming a `.swift` file and a line number** → that is the first real compile of `StarPrinterPlugin.swift`. Screenshot them and send them to me. Do not try to fix them yourself.

# Troubleshooting: "Failed Registering Bundle Identifier ... is not available"

---

If Xcode says:

```
Failed Registering Bundle Identifier
```

```
The app identifier "com.greenwaymarijuana.register" cannot be registered to
your development team because it is not available. Change your bundle
identifier to a unique string to try again.
```

**Read this before changing anything.** The instruction Apple gives you in that message — "change your bundle identifier" — is the **wrong move here**, and it is expensive. See *Do not change the bundle ID* below.

## Good news first

---

Getting this error means the previous two problems are **solved**:

- Your team is set (the panel shows **LYMAN'S MARIJUANA L.L.C.**)
- Your iPad is registered (Accounts now shows **1 Provisioned Device**)

Xcode has stopped complaining about devices entirely. It has moved on to the next step — claiming the app's name with Apple — and only that step is failing.

## What "not available" actually means

---

App IDs are globally unique across all of Apple. "Not available" means `com.greenwaymarijuana.register` **is already registered to some Apple team, and it is not the team you are currently signing with.**

The overwhelmingly likely explanation, given the order things happened: while troubleshooting the earlier "no devices" error, the **Personal Team** was selected in the Team dropdown at least once. If Xcode managed to reach Apple during that window, it claimed `com.greenwaymarijuana.register` for the **Personal Team**. Now that you have correctly switched to the paid LLC team, Apple sees the name as taken — by your *other* account.

Both accounts are yours. Nothing has been stolen or lost. The name is simply filed in the wrong drawer.

## Do not change the bundle ID

---

Apple's error message suggests changing it. **Do not.** In this project the bundle identifier is deliberately write-once:

- It is set in `src/lib/pos/capacitor-config-core.ts` as `REGISTER_APP_ID`
- `capacitor.config.ts` imports it from there
- `tests/compliance/pos-capacitor-config.test.ts` asserts its exact value
- It is written into `ios/App/App.xcodeproj/project.pbxproj` in two places

More importantly, once an app is uploaded to App Store Connect under an ID, that ID is permanent. Changing it later means a **new app record, a new App Store listing, and a reinstall on every till**. It is not worth it to dodge a five-minute fix.

Changing it is a code change in the repo, not something to edit in Xcode. If we ever genuinely need to, that is my job, not a field fix.

## Fix it — Step 1: find where the ID is registered

---

1. Go to <https://developer.apple.com/account/resources/identifiers/list>
2. **Sign in with the LLC account** (the paid one)
3. Look for `com.greenwaymarijuana.register` in the list

**If you see it there** → the LLC team already owns it. Skip to *Step 3: refresh Xcode*. This is then just a stale-cache problem on the Mac.

**If you do not see it there** → the Personal Team almost certainly owns it. Continue to Step 2.

## Fix it — Step 2: free the identifier from the Personal Team

---

***CORRECTION — read this instead.*** An earlier version of this step told you to "sign out and sign back in with your personal Apple ID." ***That advice was wrong and I have removed it.*** There is no second account to sign into. You have **one** Apple ID. It carries two teams, and you cannot log into a team — you switch between them. See the section "How do I log into my old personal account?" further down, which explains this properly and tells you what to do instead.

Skip straight to *Step 4: register it by hand on the LLC team*, and if that fails, read the section on Apple Developer Support below.

## Fix it — Step 3: refresh Xcode

---

Xcode caches signing assets aggressively, and a stale cache produces this exact error even after the underlying problem is fixed.

1. Quit Xcode completely (**Cmd-Q**)
2. Open **Terminal** and run, exactly:

```
rm -rf ~/Library/Developer/Xcode/UserData/Provisioning\ Profiles
```

This only deletes cached copies on your Mac. It deletes nothing at Apple and nothing in our project. Xcode re-downloads what it needs.

3. Reopen Xcode and the project
4. **Signing & Capabilities** → confirm **Team** is **LYMAN'S MARIJUANA L.L.C.**
5. Uncheck **Automatically manage signing**, wait a few seconds, then check it again. This forces a fresh request rather than a replay of the cached one.
6. Press **Cmd-B**

## Fix it — Step 4: register it by hand on the LLC team

---

If it still fails, create the App ID manually. You are an **Admin** on the LLC team, so you have permission.

1. Go to <https://developer.apple.com/account/resources/identifiers/list> signed in as the **LLC** account
2. Click the **+** button
3. Select **App IDs** → **Continue**
4. Select **App** → **Continue**
5. **Description:** `Greenway Point of Transaction`
6. **Bundle ID:** choose **Explicit** and type exactly:

```
com.greenwaymarijuana.register
```

Type it by hand and check it character by character. Capitalisation matters — it is all lowercase.

7. Leave every capability unchecked. We use none.
8. **Continue** → **Register**

Back in Xcode: **Signing & Capabilities** → toggle **Automatically manage signing** off and on → **Cmd-B**.

## If Apple refuses to delete or register it

---

If Apple says the identifier is in use and will not release it, stop and tell me. Do **not** change the bundle ID yourself. Contact Apple Developer Support at <https://developer.apple.com/contact/> — they can release an identifier held by an account you own. Explain that both accounts belong to you and you want the ID moved to the LLC team.

## Reminder

---

Still true, and still worth repeating: signing happens **before** compiling. My Swift printer plugin has not been compiled even once yet. Once the build clears signing, watch for red errors naming a `.swift` file and a line number — that is the real first test. Screenshot them and send them to me.

# "How do I log into my old personal account?"

---

Short answer: **you don't, because it doesn't exist**. You were right and I was wrong. You have exactly **one** Apple ID, and my last set of instructions was written as if you had two. That was my mistake, and it sent you looking for a login screen that was never there.

## One Apple ID, two teams

---

This is the single idea that makes the whole mess make sense:

- An **Apple ID** is a *login*. You have one: your `m_lyman` address.
- A **team** is a *container* that owns apps, certificates, devices and App IDs. Your one Apple ID belongs to **two** of them.

The two teams on your single Apple ID are:



| Team                                              | What it is                                                                                           | Cost |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------|------|
| <b>MICHAEL BRIAN LYMAN (Personal Team)</b>        | Created automatically by Apple the moment any Apple ID is used in Xcode. You never signed up for it. | Free |
| <b>LYMAN'S MARIJUANA L.L.C.</b><br>( DUTQ8VAG28 ) | The real one, created when you paid the \$99 and filed the business paperwork.                       | Paid |

So when you type your email and your old password and "it shows me the LLC side now" — **that is correct behaviour, not a bug**. There is one door. Both rooms are behind it. The website simply shows you the paid team because that is the one with a real membership.

**You cannot log into the Personal Team.** It has no separate password, no separate login, and on the developer website it has **no Identifiers page at all**. That is exactly why you could not find `com.greenwaymarijuana.register` to delete — not because you were looking in the wrong place, but because the place I sent you to does not exist for free teams.

You switch between teams; you do not log into them. In Xcode that switch is the **Team** dropdown in *Signing & Capabilities*. On the website it is the team name near the top of the page.

## What I verified from your six screenshots

I read all six before writing any of this. Everything below is what your LLC team account actually contains right now — team `DUTQ8VAG28`, header **Michael Lyman / LYMAN'S MARIJUANA L.L.C.**:

- **Identifiers** — contains **only** one entry: `XC Wildcard`, identifier `*`. That is a generic wildcard Xcode creates on its own; it is unrelated to us. **`com.greenwaymarijuana.register` is not on the LLC team.**
- **Devices** — one device: name `iPad`, UDID `3a62e46eb6f2b13cedaa5fa8390de0580aebfaf8`, type iPad, registered **2026/09/02**. Your iPad registration worked and is permanent.
- **Certificates** — one certificate: **MICHAEL BRIAN LYMAN**, type **Development**, platform All, expires **2027/09/02**. Valid.
- **Profiles** — **empty**. Still showing "Getting Started with Provisioning Profiles".
- **Keys** — empty. Not needed for this.
- **Services** — the standard tiles (Sign in with Apple, WeatherKit, Maps). Nothing here concerns us.

That confirms the diagnosis rather than guessing at it: the identifier is not on the paid team, the paid team is otherwise healthy, and the free team — which is holding the name — gives you no page from which to release it.

Also worth noting: the certificate is issued to **MICHAEL BRIAN LYMAN**, the exact same human name Xcode shows on the Personal Team. That is further confirmation both teams hang off your one Apple ID.

## Why "Profiles" being empty is fine

A provisioning profile is generated **automatically** by Xcode once signing succeeds. It is an *output*, not something you create first. Empty Profiles is the expected state when signing has not yet completed. Do not try to build one by hand.

## So what do I actually do now?

Your manual registration attempt was the **right instinct** and you performed it correctly. The error you got back —

```
An attribute in the provided entity has invalid value
An App ID with Identifier 'com.greenwaymarijuana.register' is not available.
Please enter a different string.
```

— is Apple confirming the name is held by a team that is not the LLC. Since that team is your own free Personal Team, and free teams expose no page to release it from, **the self-service route is genuinely exhausted**. You did not miss a button.

That leaves one correct path, and it is not a workaround:

### Contact Apple Developer Support

1. Go to <https://developer.apple.com/contact/>
2. Choose **Membership and Account** → **Bundle Identifiers / App IDs**
3. Ask for a phone call or email reply — phone is faster

4. Tell them, in plain words:

"The App ID `com.greenwaymarijuana.register` is registered to my free Personal Team. I need it released so I can register it on my paid team, LYMAN'S MARIJUANA L.L.C., team ID `DUTQ8VAG28`. Both teams are on my single Apple ID. The free team has no Identifiers page, so I cannot release it myself."

5. Have ready: your Apple ID email, the team ID `DUTQ8VAG28`, and the exact identifier `com.greenwaymarijuana.register`

This is a routine request. Support does it constantly and it is usually handled on the first call. Apple can see both teams belong to you.

**After they confirm it is released**, go back and redo *Step 4 — register it by hand on the LLC team* above, then toggle **Automatically manage signing** off and back on in Xcode and press **Cmd-B**.

## Before you call — one thing worth trying first

---

It costs two minutes and occasionally skips the phone call entirely. Xcode sometimes claims the ID successfully on its own once the local signing cache is cleared:

1. Quit Xcode completely (**Cmd-Q**)
2. In **Terminal**, run exactly:

```
rm -rf ~/Library/Developer/Xcode/UserData/Provisioning\ Profiles
```

This deletes only cached copies on your Mac. Nothing at Apple, nothing in our project.

3. Reopen Xcode and the project
4. **Signing & Capabilities** → confirm **Team** is **LYMAN'S MARIJUANA L.L.C.** — not the Personal Team
5. Uncheck **Automatically manage signing**, wait five seconds, check it again
6. Press **Cmd-B**

If the same "not available" error comes back, stop and make the support call. Do not keep retrying — it will not change on its own.

## Still do not change the bundle ID

---

Unchanged from the previous section, and worth repeating because Apple's error message actively pushes you the wrong way. `com.greenwaymarijuana.register` is written into four places in our codebase, one of which is a compliance test that asserts its exact value. Changing it is a code change I make in the repo, never a field edit in Xcode. A support call is far cheaper than a permanent rename.

## Reminder

---

Nothing here is a code problem, and nothing here is lost work. Every earlier blocker is still cleared. And signing still happens **before** compiling, so my Swift printer plugin has *still* not been compiled even once.

When the build finally clears signing, watch for red errors naming a `.swift` file and a line number. That is my code's first real test. Screenshot them and send them to me — do not fix them yourself.

## "Build Succeeded" — what to do next

---

This is the milestone. `StarPrinterPlugin.swift` **compiled for the first time**. Every previous failure was Apple paperwork; this one was my actual code, and it passed. Six blockers cleared in a row.

Two things happen now, and **they run in parallel** — do not do them in sequence:

- **Track A — test the printer on the iPad**. About 30 minutes. Do it today.
- **Track B — submit the MFi form to Star**. Ten minutes, then a **1-2 week wait**. Start it today too, because the clock does not begin until you submit.

Track B is the one people skip and regret. If you submit the app to Apple without the PPID, Apple **rejects it**, and you wait out the same two weeks anyway.

## Track A — get it running and printing

### A1. Install it on the iPad

Building is not installing. **Cmd-B** only compiles. With the iPad connected and unlocked, pick it in the destination menu and press the ► **Play** button.

First launch will likely refuse to open with an untrusted-developer message. That is normal for a first install. On the iPad:

**Settings ► General ► VPN & Device Management ►** your developer certificate ► **Trust**

Then tap the app icon again.

### A2. Pair the printer in iPadOS first

The app does not do this part; iPadOS does. If you have not already:

1. Print the self-test — printer **off**, hold **FEED**, switch it **on**, keep holding until it prints. Find the **Dev Name** line, `TSP100-XXXX`. Write it down — you need it again for the MFi form.
2. iPad **Settings ► Bluetooth** → tap that name → wait for **Connected**.

### A3. Pick the printer in the app

This is the SLICE 11 screen, which did not exist when Part 3 below was written. **Do this before running any test sale.**

In the app: **MORE ► ⌘ Receipt printer**

1. Tap **Search for printers** — it scans for six seconds
2. Your registered printer is listed **first with a ✓**, matched against the serial on its equipment record
3. Tap it to select it
4. Tap **Test print**

The test slip says **PRINTER TEST** and **"It is NOT a sale"**, names the till and printer, and shows **no dollar amount** anywhere. It also **cannot** open the cash drawer — that is enforced in code, not merely left out.

If more than one printer answers and none match the equipment record, the screen warns you and still lets you choose. That is deliberate: a warranty-replacement printer has a new serial, and refusing to print would leave the store dead. A warning is right; a locked door is not.

**If the scan finds nothing:** printer powered on, has paper, and shows **Connected** in iPadOS Bluetooth. Then scan again.

### A4. Run a real cash sale

Ring up any item, complete a **cash** sale.

**What should happen:** the receipt prints, the drawer pops, and **the screen never leaves the register**. No bounce to PassPRNT or any other app. That in-app behaviour is the entire point of this slice — if the screen jumps to another app, the printer was not selected in A3.

### A5. Confirm the drawer rules

| Do this              | Drawer should |
|----------------------|---------------|
| Cash sale            | Open          |
| Refund               | Open          |
| Void                 | Open          |
| No Sale              | Open          |
| Reprint a receipt    | Stay shut     |
| Print the day report | Stay shut     |

Reprint not opening the drawer is intentional — otherwise anyone could pop the till by reprinting an old receipt.

## A6. Test it failing — please actually do this one

Open the printer's paper lid so it physically cannot print, then ring up a cash sale.

**Expected:** the sale still completes and is recorded, and you get a clear message about the printer. **The sale must not be blocked.** If a printer fault ever stops a sale from completing, that is a bug — tell me the same day.

Then close the lid and use **MORE ► Reprint last receipt** to recover the slip.

## Track B — the MFi form (start today)

Full field-by-field instructions are in **Part 5, Step 13** above. The short version:

- Form: [https://star-m.jp/eng/support/s\\_print/app\\_regist.html](https://star-m.jp/eng/support/s_print/app_regist.html)
- Printer to tick: **TSP100III**BI — *not* TSP143IV, which is your Ethernet online-order printer
- App name: Greenway Point of Transaction
- Bundle ID: com.greenwaymarijuana.register
- Version: 1.0

Star files it with Apple and emails you an **MFi Product Plan ID (PPID)** that looks like MFI PPID 123456-7890 . **Save it somewhere permanent** — you need it for App Store submission and for every future version of the app.

## What NOT to do yet

- **Do not submit to the App Store.** Wait for the PPID. Part 6 covers submission and it is gated on that.
- **Do not turn off New Pairing Permission yet** (Step 8). Once it is off you cannot pair a second iPad without factory-resetting the printer. If a second till is coming, **pair both iPads first**.

## What to send me

Whatever happens, a short report is enough:

- Test print worked? Y/N
- Cash sale printed and popped the drawer, without leaving the register? Y/N
- Paper-lid test — did the sale still complete? Y/N
- Any red errors, or anything the app said that you did not expect — screenshot

## Reminder

The line I have repeated every round is finally retired: **the Swift plugin has now been compiled**. What has *not* happened yet is it being run against real hardware. Compiling proves the code is valid; A3 and A4 prove it actually drives your printer. Those are different things, and only you can do the second one.

# Troubleshooting: "This register cannot start" / built without a server address

If the app installs, the icon appears, you tap it, and you get:

```
This register cannot start
This copy of the register app was built without a server address, so it
cannot reach Greenway. Rebuild the app with the server address set
(REGISTER_API_BASE=https://greenwaymarijuana.com) and install it again.
Nothing is wrong with this iPad.
```

# First: this is very good news

That screen is **our own code**, not iOS. For you to be reading it, all of this had to work:

- The app was signed, installed and **trusted** by iPadOS
- It launched
- The web bundle loaded
- **My Swift plugin compiled and shipped inside the app**

Nothing crashed. The register **deliberately refused to start** because it checked itself at boot, found it had no back-office address, and stopped rather than let you discover the problem mid-sale in front of a customer. That guard lives in `src/lib/pos/register-host-core.ts` and it did exactly its job.

No sales lost, nothing to reset, nothing to undo.

## What went wrong — and it was my fault

Inside the packaged app the page is served from `capacitor://localhost`. A relative path like `/api/pos/sync` therefore resolves to *a file inside the app bundle*, which does not exist. So the server address has to be **baked into the bundle at build time**, through `REGISTER_API_BASE`.

Step 2 of this guide told you to run:

```
npm run register:build
```

It should have told you to run:

```
REGISTER_API_BASE="https://greenwaywebsite1.vercel.app" npm run register:build:ios
```

Two separate mistakes in that one line, both mine:

1. **No address.** `REGISTER_API_BASE` was missing entirely, so the bundle was built with an empty address.
2. **The wrong script.** `register:build` skips the **preflight check**. `register:build:ios` runs it — and that check exists *precisely* to catch a missing server address and stop the build **on the Mac, in one second**, before any of this. I wrote that guard, then wrote a Step 2 that walked straight around it.

I have corrected Step 2 above.

## Which address — the Vercel one

To be unambiguous, because the error message itself points at the wrong one:

| Address                                          | What it is                                                                               | Use it?      |
|--------------------------------------------------|------------------------------------------------------------------------------------------|--------------|
| <code>https://greenwaymarijuana.com</code>       | Your <b>old public WordPress site</b> . No register API on it.                           | <b>✗ No</b>  |
| <code>https://greenwaywebsite1.vercel.app</code> | The <b>new system we are building</b> . Has the register API. Verified responding today. | <b>✓ Yes</b> |

The apex domain has no `/api/pos/` endpoints at all, so a register pointed at it would install and then fail on every single call. We are still building and testing, so the registers point at Vercel.

*The error message text is wrong and that is also my bug. It hard-codes `greenwaymarijuana.com` as the example. It is only an example string inside a message, but it points at the wrong site, so it is being corrected in the same change as this section.*

Switching to the real domain later is one line and a rebuild — about ninety seconds plus reinstalling. Nothing is locked in.

## Fix it — on the Mac

```
cd ~/greenway/GREENWAY-WEBSITE
git pull
REGISTER_API_BASE="https://greenwaywebsite1.vercel.app" npm run register:build:ios
```

Watch for this line in the output:

```
OK    server address: The app will call https://greenwaywebsite1.vercel.app/api/pos/... for every request.
```

If you see `FAIL server address`, the prefix did not take. It must be on the **same line**, before `npm`, with the quotes. Do not run it as a separate command.

Then:

```
npx cap open ios
```

In Xcode, press the ► **Play** button again with the iPad connected. It reinstalls over the top — you do **not** need to delete the app first.

**Why `git pull` first:** *the corrected Step 2 and this section are now in the repo, and pulling keeps your copy of the guide in step with mine.*

## Why this could not have been caught before now

`ios/App/App/public` — the folder holding the built web bundle — is **git-ignored** (`ios/.gitignore`, line `App/App/public`). It is build output, generated on your Mac by `cap sync`, and it never reaches me. So I cannot see or test what got baked into your bundle. The preflight script is the mechanism that was supposed to protect that blind spot, and my Step 2 bypassed it.

## About "VPN & Device Management" being empty

You went to **Settings ► General ► VPN & Device Management** and found only *Add VPN Configuration* and *Sign in to Work or School Account* — no developer certificate.

**That is correct and nothing is wrong.** That entry appears only when iPadOS needs you to manually trust a developer profile. Apps installed **directly from Xcode over a cable** to a device in **Developer Mode** are trusted automatically, so no entry is created.

The proof is simple: **the app opened**. An untrusted app cannot launch at all — it shows "*Untrusted Developer*" and refuses. You got our register's own screen instead, which means trust was never in question.

So: ignore that step. It applies to ad-hoc and TestFlight-style installs, not to a cabled Xcode install. I listed it as "probably" and it turned out not to apply.

## Star MFi form — submitted ✓

Noted and logged. The 1–2 week PPID clock is now running. Nothing further to do there until Star emails the **MFi Product Plan ID**; save it when it arrives.

## What to do after the rebuild

Straight back to the printer testing from the previous section:

1. **MORE ► ⌘ Receipt printer → Search for printers →** select yours → **Test print**
2. A real **cash sale** — receipt prints, drawer pops, screen never leaves the register
3. The **paper-lid test** — open the lid, ring a cash sale, confirm the sale still completes

## Reminder

The plugin is compiled and now installed on the iPad. What still has not happened is it **driving the printer**. That is the next real test, and it needs this rebuild first.

# The printer screen said "only available on the iPad" — while running ON the iPad

You did not miss a step. **You found a real bug in my code**, and it is fixed.

## What you saw

On the iPad, in the installed app: **MORE ► ⌵ Receipt printer** opened a box saying "*Printer setup is only available on the iPad*", with nothing to tap but **Close**. Sales rang up fine, but no receipt printed and the drawer never popped. Pressing **Print receipt** gave "*the receipt window was blocked by the browser*."

Every one of those symptoms is the **same single cause**.

## The cause

The register asks one question at startup: "*is the native printer plugin here?*" If the answer is no, it falls back to browser printing — which on iOS means a pop-up, which iOS blocks. That is why you got a pop-up warning inside a native app, which should be impossible.

The answer was **wrongly** coming back "no". My code asked like this:

```
Capacitor.Plugins["StarPrinter"]
```

That looks right and it is wrong. In Capacitor 6 and later, **Capacitor.Plugins is not filled in by the native side**. It only gets filled as a side effect of calling `registerPlugin()` from JavaScript — and nothing in our code ever called it. So that lookup was **permanently empty on a perfectly healthy build**.

The native bridge does publish what it registered, in `Capacitor.PluginHeaders`, before our code runs. That is the honest source of truth, and it is what the code now reads.

**Your Swift plugin was correct and working the entire time**. It compiled, it installed, it was registered with the bridge and waiting. The JavaScript simply never asked the right question, so it was never called once.

## Why the tests did not catch it

Because every test faked the plugin by filling in `Capacitor.Plugins` — the same wrong assumption as the code. The tests and the bug agreed with each other, so 53 printing tests passed against a register that could not print.

That is the honest answer, and it is why I added **four regression tests** that fail against the old lookup and pass against the new one. I verified that by reverting the fix and watching the new test go red.

## What is fixed

- `getStarPlugin()` now checks `Capacitor.PluginHeaders` for `StarPrinter` and obtains it with `registerPlugin()`
- If the app is native but the plugin genuinely is **not** compiled in, it still returns nothing and degrades exactly as before — it does **not** hand back a broken object that would fail on every call
- The browser PWA is unchanged

## What you need to do

Rebuild and reinstall — same two commands as last time:

```
cd ~/greenway/GREENWAY-WEBSITE
git pull
REGISTER_API_BASE="https://greenwaywebsite1.vercel.app" npm run register:build:ios
npx cap open ios
```

Then ► **Play** in Xcode.

Now **MORE ► ⌵ Receipt printer** will show the real screen with a **Search for printers** button. Pair the printer in **iPadOS Settings ► Bluetooth** first if you have not, then search, select yours, and **Test print**.

## About the 10-second licence scan

Different problem, not a bug in our code, and worth stating plainly: **we have not tuned the scanner yet.**

Our timers are already near-instant — the register finalizes a scan **300 ms** after the last character arrives, with a 1200 ms safety net for stalled reads. Those are not where ten seconds comes from.

The delay is the **Socket DuraScan D760 itself**, in its default **HID keyboard mode**. In that mode it types the licence barcode into the app one character at a time, like a very fast typist. A driver's licence PDF417 holds roughly 300–1100 characters, and Socket's own documentation says Basic/HID mode is "*much slower ... for barcode symbologies encoding a lot of data, such as many 2D barcodes.*" Ten seconds for a licence is consistent with that.

There are two ways to fix it and I want your call before I build anything:

1. **Scanner-side (fast).** Change the D760's typing speed / inter-character delay with Socket's configuration barcodes. Often a large win for zero code.
2. **App-side (proper).** Use Socket's own SDK so the scan arrives as **one complete message** instead of hundreds of keystrokes. That is effectively instant, and it removes the whole class of "stray keystrokes landed in the search box" problems. It is a slice of work, not a setting.

I do not want to guess which you want, and I would rather confirm the exact D760 firmware behaviour against Socket's documentation before touching either. **Tell me to proceed and I will.**

## Order of business

Do the printer rebuild first and confirm printing works end to end. The scanner is a speed annoyance; the printer is a missing function. One at a time, so we always know which change caused which result.

## Verified before shipping

- TypeScript: **0 errors**
- Lint: **0 problems**
- **524 test files, 13,331 tests, all passing** (up from 13,327 — four new regression tests, no regressions)
- The new tests were confirmed to **fail** against the old lookup

## Reminder

The plugin compiled, installed, and is now actually reachable. What still has not happened is a receipt coming out of the printer. That is the next real test.

# Still no printer after the rebuild — the real root cause

My last fix was correct but **incomplete**, and I owe you a plain explanation of why you burned another rebuild.

## What I found

`StarPrinterPlugin.swift` **was never part of the Xcode project.**

The file has been sitting in `ios/App/App/` since SLICE 10. It was written, reviewed, committed — and never added to the App target's **Compile Sources** list. Xcode does not compile files just because they are in the folder; a file has to be a *member of the target*.

So the plugin was **never compiled into the app**. Not once. Every "Build Succeeded" you got was a build of an app that did not contain the printer code.

I verified this directly in `ios/App/App.xcodeproj/project.pbxproj` :

```
Sources build phase contained ONLY:  
AppDelegate.swift  
SceneDelegate.swift
```

No `StarPrinterPlugin.swift`. Not in the file references, not in the group, not in the build phase.

## Why last round's fix did not help

---

Last round I fixed how the JavaScript *asks* whether the plugin exists ( `Capacitor.PluginHeaders` instead of `Capacitor.Plugins` ). That fix was genuinely necessary and is correct — I re-verified it against Capacitor's own iOS source this round:

```
JSExport.swift:101  var h = (a.PluginHeaders = a.PluginHeaders || []);
```

The iOS bridge populates `PluginHeaders` and never populates `Plugins` . So the old lookup could never have worked.

But fixing the question does not help when the answer is genuinely "no". The plugin really was absent, so the honest answer was still "*not available*".

**Two independent bugs, stacked.** I found the first, fixed it, and reported success without confirming the second. That is the mistake.

## Why `npx cap sync ios` never fixed it

---

This is the part worth understanding, because it explains why no amount of rebuilding would ever have worked.

Capacitor's iOS bridge auto-registers plugins by reading `packageClassList` from `capacitor.config.json` . That list is generated by the Capacitor CLI, and the CLI builds it by scanning **installed npm plugin packages only** — `@capacitor/cli/dist/util/iosplugin.js` , `getPluginFiles()` , which walks each plugin's `rootPath` .

An **app-local** Swift file in `ios/App/App/` is outside that scan **forever**. `cap sync` will never see it, never list it, never register it.

For an app-local plugin the requirement is simply that the file is **compiled into the target**. Once it is, Capacitor's bridge finds it automatically, because it registers every compiled class conforming to `CAPPlugin` & `CAPBridgedPlugin` . Our class already declares exactly that:

```
@objc(StarPrinterPlugin)
public class StarPrinterPlugin: CAPPlugin, CAPBridgedPlugin {
    public let jsName = "StarPrinter"
```

So target membership was the **only** thing missing.

## What I changed

---

I added the file to the Xcode project properly — file reference, navigator group, and critically the **Sources build phase**, which is the entry that actually causes compilation.

I also added three compliance tests that read `project.pbxproj` and fail if the plugin ever falls out of the build phase again. I confirmed they work by reverting the project file and watching them go red.

**This is now caught by the test suite instead of by you, at the counter.**

## What you need to do

---

Same commands. Nothing new to click in Xcode — the project file itself is fixed, so pulling brings the fix with it.

```
cd ~/greenway/GREENWAY-WEBSITE
git pull
REGISTER_API_BASE="https://greenwaywebsite1.vercel.app" npm run register:build:ios
npx cap open ios
```

Then press ► **Play**.

### ⚠ This build is the first real compile of my Swift code

Everything up to now compiled an app **without** that file in it. This is the first time the compiler will actually read `StarPrinterPlugin.swift` .

**Red errors naming `StarPrinterPlugin.swift` and a line number are now possible, and they are not a disaster.** They would be ordinary, mechanical mistakes — a renamed SDK method, a changed argument label. Screenshot them and send them to me. **Do not try to fix them yourself.**

If it builds clean, then **MORE ► ↗ Receipt printer** will finally show the real screen with a **Search for printers** button.

## How you will know it worked

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The printer screen is the tell:

- **Still says "only available on the iPad"** → the plugin still is not registering. Send me a screenshot; do not rebuild repeatedly.
- **Shows "Search for printers"** → the plugin is compiled, registered, and reachable. Pair in **iPadOS Settings ► Bluetooth** first, then search, select yours, and **Test print**.

## Verified before shipping

---

- TypeScript: **0 errors**
- Lint: **0 problems**
- **524 test files, 13,334 tests passing** (up from 13,331 — three new tests, no regressions)
- New tests confirmed to **fail** against the unpatched project file
- Capacitor bridge behaviour confirmed against `@capacitor/ios` source, not assumed

## Socket scanner — approved, queued as its own slice

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You chose the **app-side SDK** route, which is the right call. That is a proper slice and I will not bolt it onto this one; mixing a scanner rewrite into an unverified printer fix would make it impossible to tell which change caused which result.

**Printer first.** Once a receipt physically prints, I will start the Socket slice: research the DuraScan D760's SDK against Socket's official documentation, then replace HID keystroke-streaming with a single delivered scan payload — which also eliminates the stray-keystroke class of bugs entirely.